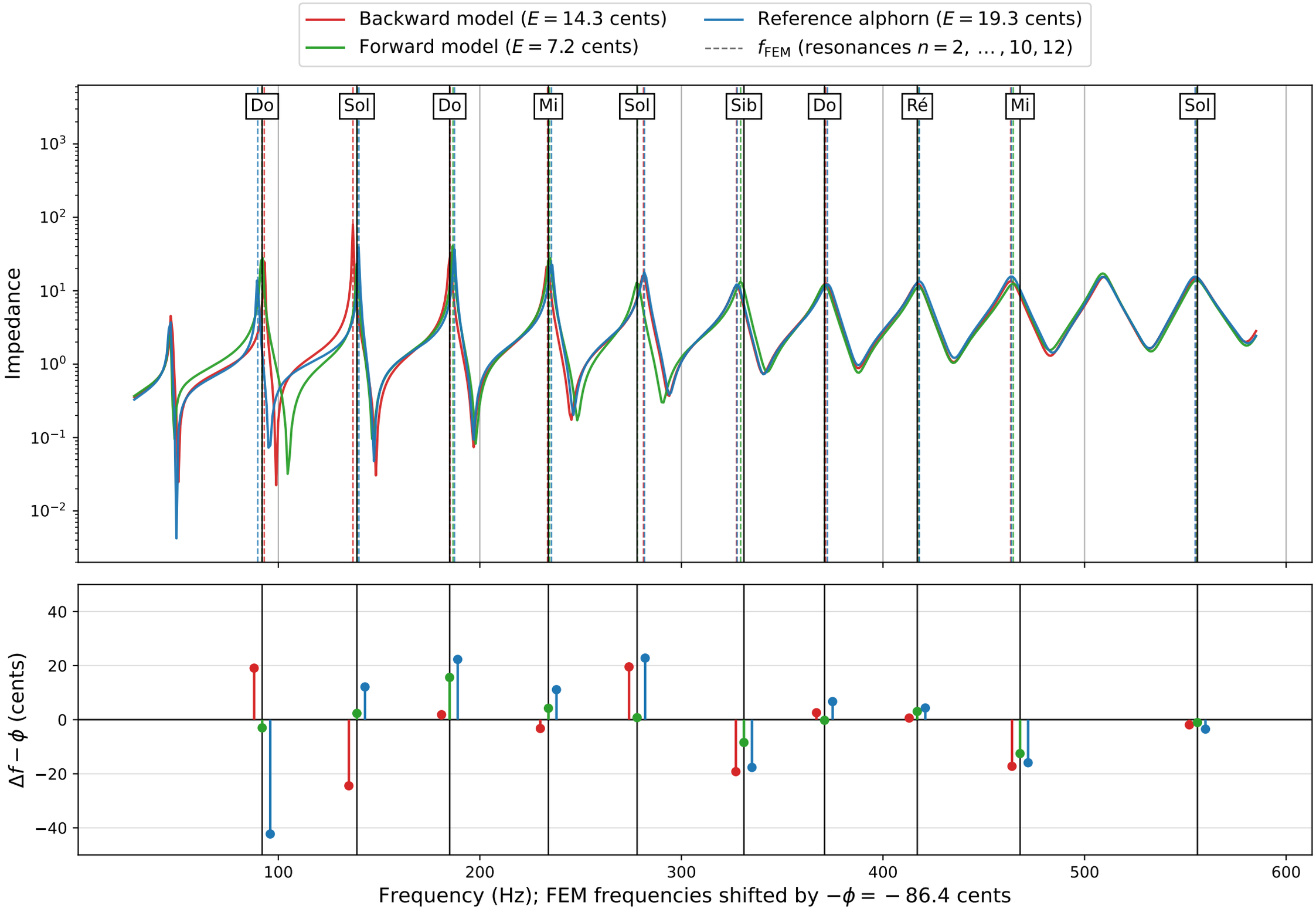
Backward model ($E = 14.3$ cents)
Forward model ($E = 7.2$ cents)
Reference alphorn ($E = 19.3$ cents)
f_{FEM} (resonances $n = 2, \ldots, 10, 12$)
Do
Sol
Do
Mi
Sol
Sib
Do
Ré
Mi
Sol
Impedance
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
$\Delta f - \phi$ (cents)
40
20
0
−20
−40
100
200
300
400
500
600
Frequency (Hz); FEM frequencies shifted by $-\phi = -86.4$ cents